816 - b)

$$\int (2x^2-3x)\sin\left(\frac{x}{3}\right)dx = \left| \begin{matrix} u=(2x^2-3x) & v'=\sin\left(\frac{x}{3}\right) \\ u'=4x-3 & v=-3\cos\left(\frac{x}{3}\right) \end{matrix} \right| \overset{PP}{=} (2x^2-3x)\cdot\left(-3\cos\left(\frac{x}{3}\right)\right) - \int (4x-3)\cdot\left(-3\cos\left(\frac{x}{3}\right)\right)dx =$$

$$= (9x-6x^2)\cos\left(\frac{x}{3}\right) - \int (9-12x)\cos\left(\frac{x}{3}\right)dx = \left| \begin{matrix} u=(9-12x) & v'=\cos\left(\frac{x}{3}\right) \\ u'=(-12) & v=3\sin\left(\frac{x}{3}\right) \end{matrix} \right| \overset{PP}{=}$$

$$\overset{PP}{=} (9x-6x^2)\cos\left(\frac{x}{3}\right) - \left((9-12x)\cdot 3\sin\left(\frac{x}{3}\right) - \int (-12)\cdot 3\sin\left(\frac{x}{3}\right)dx\right) =$$

$$= (9x-6x^2)\cos\left(\frac{x}{3}\right) - \left((27-36x)\cdot\sin\left(\frac{x}{3}\right)\right) + \int -36\sin\left(\frac{x}{3}\right)dx =$$

$$= (9x-6x^2)\cos\left(\frac{x}{3}\right) - (27-36x)\cdot\sin\left(\frac{x}{3}\right) + 108\cos\left(\frac{x}{3}\right) + C = \underline{(-6x^2+9x+108)\cos\left(\frac{x}{3}\right) + (36x-27)\sin\left(\frac{x}{3}\right) + C}$$

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914 - h)

$$\int \frac{-x^2-15x+25}{(x+1)(x^2-4x+8)}dx = \int\left(\frac{A}{(x+1)} + \frac{Bx+C}{x^2-4x+8}\right)dx = \int\left(\frac{3}{x+1} + \frac{-4x+1}{x^2-4x+8}\right)dx$$

$$A = \left.\frac{-x^2-15x+25}{x^2-4x+8}\right|_{x=-1} = \frac{-1+15+25}{1+4+8} = \frac{39}{13} = 3$$

$$-x^2-15x+25 = A(x^2-4x+8) + (Bx+C)\cdot(x+1) = Ax^2-4Ax+8A+Bx^2+Bx+Cx+C =$$

$$= (A+B)x^2 + (-4A+B+C)x + 8A+C$$

$$\left.\begin{matrix} -1 = A+B \\ -15 = B+C-4A \\ 25 = 8A+C \end{matrix}\right\} \Rightarrow \begin{matrix} -1 = 3+B \rightarrow B=-4 \\ -15 = B+C-12 \\ 25 = 24+C \rightarrow C=1 \end{matrix} \quad \left.\begin{matrix} -15 = -4+1-12 \\ -15 = -15 \end{matrix}\right\}$$

$$= \int\left(\frac{3}{x+1} + \frac{-2\cdot(2x-4)-7}{x^2-4x+8}\right)dx = \int\left(\frac{3}{x+1} - 2\frac{2x-4}{x^2-4x+8} - 7\frac{1}{(x-2)^2+4}\right)dx =$$

$$= \underline{3\ln|x+1| - 2\ln(x^2-4x+8) - \frac{7}{2}\operatorname{arctg}\frac{x-2}{2} + C} \quad \text{na } \mathbb{R}-\{-1\}$$

10/1-c)

$$\int \sin^4 x \cos^2 x \, dx = \cancel{\int \sin^4 x (1-\sin^2 x) dx = \int (\sin^4 x - \sin^6 x) dx} =$$

$$= \int \left(\frac{1-\cos 2x}{2}\right)^2 \cdot \frac{1+\cos 2x}{2} dx = \frac{1}{8}\int (1-\cos 2x)^2 \cdot (1+\cos 2x) dx = \cancel{\frac{1}{8}\int (1-(\cos^2 x - \sin^2 x))^2 \cdot (1+\cos^2 x - \sin^2 x) dx}$$

$$= \cancel{\frac{1}{8}\int (2\sin^2 x)^2 \cdot 2\cos^2 x \, dx = \frac{1}{8}\int 2(2\sin^4 x \cos^2 x)}$$

$$= \frac{1}{8}\int (1-\cos 2x)^2 \cdot (1+\cos 2x) dx = \left| \begin{matrix} t = 2x \\ dt = 2dx \end{matrix} \right| =$$

$$= \frac{1}{16}\int (\cos^3 t - \cos^2 t - \cos t + 1) dt = \frac{1}{48}\sin(t)\cos^2(t) - \frac{1}{48}\int \cos t \, dt -$$

$$- \frac{1}{16}\int \cos^2 t \, dt + \frac{1}{16}\int 1 \, dt = -\frac{\sin t}{48} + \frac{1}{48}\sin t \cos^2 t - \frac{1}{16}\int \left(\frac{1}{2}\cos 2t + \frac{1}{2}\right) dt +$$

$$\frac{1}{16}\int 1 \, dt = -\frac{\sin t}{48} + \frac{1}{48}\sin t \cos^2 t - \frac{1}{32}\int \cos 2t \, dt + \frac{1}{32}\int 1 \, dt = \left| \begin{matrix} s = 2t \\ ds = 2dt \end{matrix} \right| =$$

$$= \frac{-\sin t}{48} + \frac{\sin t \cos^2 t}{48} - \frac{\sin(s)}{64} + \frac{t}{32} + c = \frac{-\sin t}{48} + \frac{\sin t \cos^2 t}{48} -$$

$$- \frac{\sin 2t}{64} + \frac{t}{32} + c = \frac{-\sin(2x)}{48} + \frac{\sin(2x)\cos^2(2x)}{48} - \frac{\sin 4x}{64} + \frac{x}{16} + c$$

$$= \frac{x}{16} - \frac{\sin 4x}{64} - \frac{\sin(2x)}{48} + \frac{\sin(2x) - \sin^3(2x)}{48} + c = \underline{\underline{\frac{x}{16} - \frac{\sin 4x}{64} - \frac{\sin^3 2x}{48} + c}} \text{ na } \mathbb{R}$$

10/2 - a)

$$\int \frac{3e^{4x}-7e^{2x}}{e^{4x}-4e^{2x}+3} = \left| \begin{array}{l} e^{2x}=u \\ x=\frac{1}{2}\ln u \\ dx=\frac{1}{2u}\,du \end{array} \right| = \frac{1}{2}\int \frac{3u^2-7u}{u^2-4u+3}\cdot\frac{1}{u}\,du =$$

$$= \frac{1}{2}\int \frac{u\cdot(3u-7)}{u\cdot(u^2-4u+3)}\,du = \frac{1}{2}\int \frac{3u-7}{u^2-4u+3}\,du = \frac{1}{2}\int \frac{3u-7}{(u-1)\cdot(u-3)}\,du =$$

$$\frac{A}{u-1}+\frac{B}{u-3}$$

$$A = \frac{3u-7}{u-3}\Big|_{u=1} = \frac{-4}{-2} = 2 \qquad B = \frac{3u-7}{u-1}\Big|_{u=3} = \frac{2}{2} = 1$$

$$= \frac{1}{2}\int \frac{2}{(u-1)}+\frac{1}{(u-3)}\,du = \ln|u-1| + \frac{\ln|u-3|}{2} + C =$$

$$= \ln|e^{2x}-1| + \frac{\ln|e^{2x}-3|}{2} + C \text{ na } R-\left\{0;\ \frac{\ln 3}{2}\right\}$$

1013 - a)

$$\int \frac{\ln x + 3}{x(\ln^2 x + 4\ln x + 5)}\,dx = \left| \begin{matrix} \ln x = u \\ \frac{1}{x}dx = du \end{matrix} \right| = \cancel{\frac{u+3}{u^2+4u+5}} \int \frac{u+3}{u^2+4u+5}\,du = \cancel{\int \frac{u+3}{}}$$

$$= \frac{1}{2}\int \frac{2u+4}{u^2+4u+5}\,du + \int \frac{1}{(u+2)^2+1}\,du = \left| \begin{matrix} u^2+4u+5 = s \\ (2u+4)du = ds \end{matrix} \right| = \frac{1}{2}\cdot \ln s + \int \frac{1}{(u+2)^2+1}\,du =$$

$$= \left| \begin{matrix} (u+2) = p \\ du = dp \end{matrix} \right| = \frac{\ln s}{2} + \int \frac{1}{p^2+1}\,dp = \frac{\ln s}{2} + \operatorname{arctg} p + C = \frac{\ln(u^2+4u+5)}{2} + \operatorname{arctg}(u+2) + C =$$

$$= \frac{\ln(\ln^2(x) + 4\ln x + 5)}{2} + \operatorname{arctg}(\ln x + 2) + C \quad \text{na } (0; \infty)$$

1014 - c)

$$\int \frac{\sin^2 x + 4\cos x \sin x}{5\cos^4 x + 4\cos^3 x \sin x + \cos^2 x \sin^2 x}\,dx = \cancel{\int \frac{\sin x(\sin x + 4\cos x)}{\cos^2 x(5}}$$

$$= \int \frac{\sin x(\sin x + 4\cos x)}{\cos^2 x(5\cos^2 + 4\cos x \sin x + \sin^2 x)}\,dx = \int \frac{\operatorname{tg} x(\sin x + 4\cos x)}{\cos x(5\cos^2 + 4\cos x \sin x + \sin^2 x)}\,dx =$$

$$= \int \frac{\operatorname{tg} x(\operatorname{tg} x + 4)}{5\cos^2 + 4\cos x \sin x + \sin^2 x}\,dx = \int \frac{\operatorname{tg}^2 x + 4\operatorname{tg} x}{5\cos^2 + 4\cos x \sin x + \sin^2 x}\,dx$$

1014 - c)

$$\int \frac{\sin^2 x + 4\cos x \sin x}{5\cos^4 x + 4\cos^3 x \sin x + \cos^2(x)\sin^2 x}\,dx = \cancel{\int \frac{\sin x(\sin x + 4\cos x)}{\cos^2 x(5\cos^2 x + 4\cos x \sin x + \sin^2 x)}} =$$

$$= \int \frac{\operatorname{tg} x(4+\operatorname{tg} x)\arccos^2 x}{5+4\operatorname{tg} x+\operatorname{tg}^2 x}\,dx = \left| \begin{matrix} u = \operatorname{tg} x \\ du = \arccos^2 x\,dx \end{matrix} \right| = \int \frac{u(4+u)}{5+4u+u^2}\,du =$$

$$= \int \left(1 - \frac{5}{u^2+4u+5}\right) du = -5\int \frac{1}{u^2+4u+5}\,du + \int 1\,du = -5\int \frac{1}{(u+2)^2+1}\,du + \int 1\,du =$$

$$= \left| \begin{matrix} s = u+2 \\ ds = du \end{matrix} \right| = -5\int \frac{1}{s^2+1}\,ds + \int 1\,du = -5\operatorname{arctg}(s) + u + C = u - 5\operatorname{arctg}(u+2) + C =$$

$$= \operatorname{tg} x - 5\operatorname{arctg}(\operatorname{tg} x + 2) + C \quad \text{na } \left(-\frac{\pi}{2} + k\pi; \frac{\pi}{2} + k\pi\right) k \in \mathbb{Z}$$

1017 - b)

$$\int \frac{\sqrt[4]{x+2} - 4}{\sqrt{x+2} - 4\sqrt[4]{x+2} + 4}\,dx = \cancel{\int \frac{\sqrt[4]{x+2} - 4}{(\sqrt[4]{x+2} - 2)^2}\,dx} \left| \begin{matrix} u = \sqrt{x+2} \\ du = \frac{1}{2\sqrt{x+2}}\,dx \end{matrix} \right| = 2\int -\frac{u(\sqrt[4]{u^2} - 4)}{4\sqrt[4]{u^2} - u - 4}\,du =$$

$$= \left| \begin{matrix} p = \sqrt{u} \\ dp = \frac{1}{2\sqrt{u}}\,du \end{matrix} \right| = -4\int -\frac{(p-4)p^3}{(p-2)^2}\,dp = 4\int \left(p^2 - \frac{16}{p-2} - \frac{16}{(p-2)^2} - 4\right) dp =$$

$$= \frac{4p^3}{3} - 16 + \frac{64}{p-2} - 64\log(p-2) + C = \frac{4}{3}u^{\frac{3}{4}} - 16\sqrt[4]{u} + \frac{64}{\sqrt[4]{u}-2} - 64\ln|\sqrt{u} - 2| + C =$$

$$= \frac{4}{3}(x+2)^{\frac{3}{4}} - 16\sqrt[4]{x+2} + \frac{64}{\sqrt[4]{x+2} - 2} - 64\ln|\sqrt[4]{x+2} - 2| + C \quad \text{na } (-2; 14) \cup (14; \infty)$$